Response to Reviewer FBhK

BioCoLoop, submission 5046 | Revision dated 11 November 2026.

Thank you for the close reading. We are glad that the practical problem, the common interface across task families and the appendix detail were useful, and that the budget-matched allocation result (+10.84 Top-1, 95% CI [4.54, 17.73]) read as a positive finding. Your two questions led to the main additions in this revision. In brief: at matched access, history-guided proposals help on some tasks but are not consistently better than direct, random or GP-EI search. Collaboration improves all five endpoint means, and collaborative design search and training allocation add task-specific gains. The revision reports both sides.

Table numbers changed in the revision. Your Tables B.10, B.12 and B.13 are now Tables B.9(a), B.9(b) and B.10.

1. Added value of the loop beyond collaborative fitting, and random/direct search at the same access (Weakness 1, Question 1)

Collaborative fitting versus the loop. We added the missing control: the fixed default design (D00) trained by all ten laboratories at the main seeds 42-44 (Appendix Table D.1; the D00 arm of Table A.1). Relative to it, the ten-laboratory loop adds 1.96 AUROC on DTI (93.76 vs 91.80) and 6.77 Top-1 points on Norman (22.25 vs 15.48), where the default design degrades as laboratories are added. It nearly matches the fixed model on proteomics (36.99 vs 37.07) and ties it on VCC (31.54 for both) and is 3.48 points lower on Tahoe. On DTI, proteomics and VCC, most of the gain from one to ten laboratories therefore comes from shared data; collaborative design search adds to it on DTI and Norman. Section 4.3 now states this decomposition.

Same laboratories, panels, library and budget. Appendix E.1-E.2 runs random search without replacement (the exact expectation over all 462 six-design subsets), GP-EI, and direct and history-guided Qwen and Luna. All share the ten laboratories, development panels, twelve configurations, six proposal slots and 100-round fits, and failed requests are charged. This is a single-seed study per task (DTI seed 42, others seed 61) replayed from a fixed candidate cache. Results: - Norman: Luna history 23.11 Top-1, direct 22.30, random 22.66 (27 test interventions). - DTI: direct Luna and GP-EI 94.60, history 94.27. - Proteomics: all policies tie. - VCC: language-model searches and GP-EI 31.50, random 27.53.

Two follow-ups use 20-round fits under separate protocols, summarized in Section 4.7: E.6 uses four gene-disjoint study-cell contexts, and E.9 uses 36 wider recipes, a shared historical seed-42 starting model and pooled development selection. Both originally had three seeds; for the revision we extended each to eight, with the analysis fixed before the new fits and no seeds excluded: - E.6, gene-disjoint cells: structured history 22.75 Top-1, direct 22.34, random 22.69, GP-EI 23.28, fixed design 21.37. Structured history beats direct by +0.41 (seed-block 95% CI [0.09, 0.84]; four wins, four ties, no losses), is 0.06 points above random in mean (the registered primary contrast; CI [-0.33, 0.46], four wins and four losses, so no clear advantage) and trails GP-EI (-0.52). - E.9, 36 wider recipes (Luna, eight seeds): on DTI, structured history reaches 94.37 AUROC against 94.32 for random (+0.053, CI [0.050, 0.055], eight wins of eight) and 94.29 for GP-EI; direct search selects the same designs. On protein, history is +0.21 AP above random (CI [-0.04, 0.44]; six wins, two losses), while direct search (38.99) and GP-EI (39.02) are higher. Qwen history is above random on DTI (+0.04) and below it on protein (-0.44). - Two retrospective E.9 checks without new fits: most Qwen failures listed the changed fields incorrectly for a valid design; canonicalizing that redundant metadata cuts invalid requests from 102 to 5 of 288, after which Qwen selects the same DTI predictors as Luna in every seed and its direct search reaches 38.98 protein AP (random 38.58, GP-EI 39.02). Replacing the pooled development score by laboratory-level selection rules lowers protein AP in all eight seeds, so pooling evidence across laboratories is the better selection signal.

A separate source-defined study (E.4; seeds 61-63, 100-round fits) finds no search policy exceeding the fixed design.

So at least one tested language-model policy exceeds the random expectation in mean in each family; the successful policy varies by task, and history has no consistent advantage over GP-EI or random search. In the separate 36-design, 24-slot study, final history and direct scores tie on seven of eight comparisons; history is lower on VCC/Luna (20.01 versus 20.23). Section 4.7 and Appendix B.3 now report the original eight ties, one gain and one loss in the main text as well.

Efficiency and allocation. The target crossings in Table B.10 are retrospective trajectory diagnostics: every scheduled proposal was executed, so they do not imply saved computation, and the revision now describes them that way. For allocation, a fresh 360-fit study (Table E.5) adds random and score-only promotion. Trajectory promotion beats uniform allocation in four task-seed pairs and loses none, and it beats score-only promotion in four pairs and loses one. Against random promotion it wins six pairs and loses six (mean -0.44). The robust ingredient is screening on early collaborative evidence and then training a few candidates longer; Section 4.4 now says this.

2. Development-test divergence in the extended VCC and Tahoe searches (Weakness 2, Question 2)

Panel construction and weighting. Development and test use disjoint interventions: VCC has 60/20/20 training/development/test interventions, and Tahoe has 180/60/67. The 20 VCC development interventions form ten panels of two, and the 60 Tahoe compounds form ten panels of six. Every panel holds the same number of interventions and Top-1 is macro-averaged over interventions, so equal panel weighting equals the pooled intervention-macro score exactly. Laboratory weighting and query counts therefore do not explain the divergence.

Audit. We reconstructed all 14 selected checkpoints and their 140 development panels, and we reproduced all 80 archived held-out prefix scores (Appendix E.3). The scores reproduce correctly. The roster audit below shows that the development advantage is sensitive to distractor choice, consistent with unstable selection on a small development population; biological development-test shift may also contribute. For example, Tahoe's +5.00 development points correspond to three more correct development compounds, while held-out Top-1 falls from 25.37 to 13.43.

Roster sensitivity. We re-scored the same initial and final checkpoints on 20 paired development-roster redraws without retraining. Qwen/VCC's +1.17 development advantage becomes -1.34 ± 7.37 (positive in 6/20 draws), and Tahoe's +5.00 becomes -1.33 ± 4.82 (7/20). Luna/VCC stays positive in 12/20 draws. The large held-out declines thus coincide with development gains that depend on the choice of distractors.

Retrospective stability-rule replay. We froze three stricter acceptance rules before this replay's held-out rescoring and applied them to the original fixed proposal traces (Appendix E.3), reusing previously exposed test sets: - the gain must exceed its across-laboratory SD; - at least 6 of 10 laboratories must improve; - both laboratory halves must improve.

All three keep the default design on Tahoe (+3.48 Top-1) but also reject the helpful Norman design (-6.77). The net change is near zero (majority rule: three wins, nine ties, three losses; mean -0.64, 95% CI [-1.28, 0.02]). We therefore keep the original rule and report the trade-off. Testing larger development populations is a natural next step for stabilizing selection.

Other grouped partitions. Appendix E.6 uses four Replogle/Nadig study-cell contexts with gene-disjoint partitions (1,208/414/406 training/development/test genes). In the original three seeds every final search exceeds the fitted default (21.43 Top-1), and we report the one small interim reversal (seed 62, slot 2: +0.057 development, -0.257 test). In the eight-seed extension, structured history exceeds the fixed design in every seed, while direct search falls below it once (seed 67).

Thank you again; both questions changed what the paper now reports.

Response to Reviewer AfJB

BioCoLoop, submission 5046 | Revision dated 11 November 2026.

Thank you for the expert review, for pointing us to The Virtual Biotech, AutoScientists and Helmsman, and for your note on the matched-training-budget allocation experiment. We agree with your reading of the single-laboratory tables and of Section 4.7. The revision does not dispute these results. It clarifies the method's purpose: enable laboratories that cannot pool data to develop a predictor together, and measure the benefit relative to developing it alone. It also adds the control needed to separate shared data from collaborative design search.

Tables A.1-A.3 keep their numbers. The ten comparisons you cite from Section 4.7 are reported in Section 4.7 and in Appendix B.3.

1. Methodological novelty relative to The Virtual Biotech, AutoScientists and Helmsman

The revised related work discusses all three. In The Virtual Biotech and AutoScientists the collaborators are AI agents working on data they can access, and they share evidence, programs and experiment logs. In BioCoLoop the collaborators are data-holding laboratories. Each trains candidates on its own data and scores them on its own development panels, and only model updates and bounded per-laboratory evidence leave the site. Helmsman is directly related prior work: its agents synthesize and debug federated-learning system code in simulation. BioCoLoop instead keeps the federated protocol fixed and uses the laboratories' evidence to decide which predictor design to keep, what to try next and which candidates to train longer.

The revision also discusses federated hyperparameter search (FedEx), architecture search (FedNAS) and the FedHPO-Bench benchmark. FedEx and FedNAS optimize a predefined numerical space with a fixed algorithm; BioCoLoop contributes a common proposal, fitting and development-evidence interface that supports different biological task models and research agents. As a check of that interface, a seed-42 pilot in which AI-Scientist-v2 drives BioCoLoop's ten-laboratory interface (Appendix C.4) reaches 95.07 versus 94.60 DTI AUROC and matches BioCoLoop on the other four endpoints.

We do not claim the first agentic federated framework. The contributions we claim are: - the collaborative model-development setting and its interface across three biological task families; - early-evidence allocation, which improves Norman Top-1 by 10.84 points (95% CI [4.54, 17.73]) at matched training and evaluation budgets; - controlled experiments that separate laboratory participation, the proposal policy and training allocation.

2. With one laboratory, BioCoLoop is essentially identical to direct optimization (Tables A.1-A.3)

Your reading is correct. Appendix D.1 now explains it. With one laboratory, every strategy, BioCoLoop included, selects designs on laboratory 0's small development panel alone. The designs selected differ across strategies and seeds but score similarly, and on proteomics and the cell tasks they fall below the single-laboratory fixed model (for example, Norman 14.15 for BioCoLoop versus 25.56 fixed), and their local-development rankings do not transfer reliably to the held-out set. At ten laboratories, the main direct and history-guided workflows attain the same primary score in every run, although their selected designs can differ. History is therefore not what separates the rows in these tables.

What separates them is collaboration. Changing only participation from one laboratory to ten, the same workflow moves: - DTI AUROC from 89.89 to 93.76; - proteomics AP from 20.13 to 36.99; - VCC, Norman and Tahoe Top-1 from 20.22, 14.15 and 15.42 to 31.54, 22.25 and 22.39.

All five improve in mean (Appendix D.2). The three cell intervals against single-laboratory random search span zero, and Norman and Tahoe final Top-1 remain close to the 20% chance level. The secondary endpoints follow the same general pattern.

We also added a ten-laboratory fixed model (Appendix Table D.1) to separate shared data from design search. Relative to it, BioCoLoop adds 1.96 AUROC on DTI and 6.77 Top-1 on Norman, where the default design falls from 25.56 to 15.48 as laboratories are added. It nearly matches on proteomics (36.99 versus 37.07 AP), ties on VCC and is 3.48 lower on Tahoe. On DTI, proteomics and VCC, most of the collaboration gain is thus shared data; design search adds to it on DTI and Norman. Laboratory identity matters as well. A seed-42 fixed-model sensitivity over laboratory identities (Appendix E.7, Table E.9) places laboratory 0 below the average on proteomics and VCC; the mean VCC score across the ten single-laboratory fits is 32.50, compared descriptively with the main three-seed ten-laboratory mean of 31.54.

3. Table 1 gives BioCoLoop ten laboratories while the AI-scientist baselines use one

This is intended, and the revision now labels it clearly. Table 1 compares collaborative development with a laboratory working alone, which is the usual comparison in federated learning (cited in Section 4.2). Its caption now says that rows at the two access levels compare complete scenarios, not search methods. Section 4.3 explains why the reference row is the single-laboratory fixed model rather than the single-laboratory BioCoLoop run used elsewhere: the fixed model is the stronger local baseline (Norman 25.56 versus 22.25 for ten-laboratory BioCoLoop). Two additional controls separate collaborative fitting from design search and test another controller at matched access: - the ten-laboratory fixed model above; - the seed-42 pilot (Appendix C.4), which gives both agents the same ten laboratories. AI-Scientist-v2 matches BioCoLoop on four endpoints and reaches 95.07 versus 94.60 DTI AUROC; repaired AI-Researcher matches on proteomics, VCC and Norman, with DTI and Tahoe unscored. This supports the interface's portability across controllers.

4. In Section 4.7, eight of ten comparisons show identical final performance

We agree, and Section 4.7 now states this in its first paragraph. The original same-access study (single seed) has eight ties, a Norman gain for Luna history (+0.81) and a DTI loss (-0.33). In the separate 36-design, 24-slot study, final history and direct scores tie on seven of eight comparisons; history is lower on VCC/Luna (20.01 versus 20.23).

Two follow-ups use 20-round fits under separate protocols: E.6 (gene-disjoint study-cell contexts) and E.9 (36 wider recipes, a shared historical seed-42 start and pooled development selection). For the revision we extended both from three to eight seeds, with the analysis fixed before the new fits and no seeds excluded: - E.6: structured history reaches 22.75 Top-1 against 22.34 for direct (paired +0.41, seed-block 95% CI [0.09, 0.84]; four wins, four ties, no losses) and 21.37 for the fixed design (eight wins of eight). Against random search, the registered primary contrast, it is +0.06 in mean (CI [-0.33, 0.46]; four wins, four losses), so no clear advantage; it trails GP-EI (23.28). - E.9 (Luna): on DTI, structured history beats random by +0.053 AUROC (CI [0.050, 0.055]; eight of eight seeds) and GP-EI in every seed, but direct search selects the same designs, so history adds nothing there. On protein, history is +0.21 AP above random (CI [-0.04, 0.44]; six wins, two losses) and below direct search (-0.20, CI [-0.42, -0.01]) and GP-EI. Every history search exceeds the fitted starting design in all eight seeds. - Two retrospective E.9 checks without new fits: most Qwen failures listed the changed fields incorrectly for a valid design; canonicalizing that redundant metadata cuts invalid requests from 102 to 5 of 288, after which Qwen selects the same DTI predictors as Luna in every seed and its direct search reaches 38.98 protein AP (random 38.58, GP-EI 39.02). Replacing the pooled development score by laboratory-level selection rules lowers protein AP in all eight seeds, so pooling evidence across laboratories is the better selection signal. - E.4 (seeds 61-63, 100-round fits, source-defined laboratories): no search policy exceeds the fixed design.

At least one tested language-model policy exceeds the random expectation in mean in each family, but the successful policy varies by task. History is therefore one task-dependent component: where the evidence is clearest it improves on feedback-free proposals, but it is not consistently better than GP-EI or random search. The paper no longer presents it as the source of the main gain.

5. Allocation

In a fresh 360-fit study (Appendix E.5), trajectory promotion beats uniform allocation in four task-seed pairs and loses none. It beats score-only promotion four times and loses once, and it splits six-six against random promotion (mean -0.44). On Tahoe, trajectory promotion reaches 22.89 versus 20.90 for score-only and 21.95 for random, a task-specific benefit of the trajectory rule. The robust ingredient is screening candidates on early collaborative evidence and then training a few of them longer. The abstract and Section 4.4 now describe the result this way.

Thank you again; your comments made the paper's claims clearer.

Response to Reviewer cR5g

BioCoLoop, submission 5046 | Revision dated 11 November 2026.

Thank you for the review and for highlighting the setting we target: distributed training in medicine and biology when laboratories cannot share data. You are right that the original submission lacked random search and that its laboratories were simulated. The revision adds random, Bayesian and exhaustive search for one laboratory, random and Bayesian controls at matched ten-laboratory access, and a ten-laboratory fixed model. It also describes the simulated laboratories more precisely and reports the source-defined experiments with every result, favorable or not.

1. No comparison to traditional hyperparameter tuning or random search

One laboratory. Appendix D runs random search, GP-EI Bayesian optimization and exhaustive enumeration of the configuration menu on laboratory 0. None reaches the collaborative result. For example: - DTI AUROC: 90.10 for the conventional controls versus 93.76 for ten-laboratory BioCoLoop. - Proteomics AP: 20.13 for random search versus 36.99.

On the cell endpoints the gaps are smaller (chance is 20%), and their intervals against single-laboratory random search span zero.

Ten laboratories. Appendix E.1-E.2 compares random search without replacement (the exact expectation over all 462 six-design subsets), GP-EI, and direct and history-guided language-model search. All use the same laboratories, panels, configuration library and proposal budget, and failed requests are charged. This study replays a fixed candidate cache with one training seed per task: 42 for DTI and 61 otherwise. Conventional search is competitive: - Proteomics: all policies tie. - DTI: GP-EI and direct Luna 94.60, history 94.27. - Norman: Luna history 23.11 Top-1, direct 22.30, random 22.66 (27 test interventions).

Two follow-ups use separate protocols: E.6 uses gene-disjoint study-cell contexts and 20-round fits; E.9 uses 36 wider recipes, 20-round warm-start fits and pooled selection. We extended both from three to eight seeds for the revision, with the analysis fixed before the new fits: - E.6: GP-EI 23.28, structured history 22.75, random 22.69, direct 22.34. History beats direct (+0.41, seed-block 95% CI [0.09, 0.84]); against random it is +0.06 (CI [-0.33, 0.46]), with no clear advantage, and it trails GP-EI. - E.9 (Luna): DTI structured history 94.37 AUROC, direct 94.37, random 94.32, GP-EI 94.29; history beats random by +0.053 (CI [0.050, 0.055]) in all eight seeds. Protein: GP-EI 39.02 AP, direct 38.99, history 38.79, random 38.58; history versus random is +0.21 (CI [-0.04, 0.44]; six wins, two losses). - Two retrospective E.9 checks without new fits: most Qwen failures listed the changed fields incorrectly for a valid design; canonicalizing that redundant metadata cuts invalid requests from 102 to 5 of 288, after which Qwen selects the same DTI predictors as Luna in every seed and its direct search reaches 38.98 protein AP (random 38.58, GP-EI 39.02). Replacing the pooled development score by laboratory-level selection rules lowers protein AP in all eight seeds, so pooling evidence across laboratories is the better selection signal.

At least one tested language-model policy exceeds the random expectation in mean in each family; the successful policy varies by task, margins over random are small, and the ranking against GP-EI depends on the task. Section 4.7 says so.

2. The laboratories are random splits of one dataset, not independent entities with domain shift

We agree that the main laboratories are simulated from one dataset per task, as in most federated-learning benchmarks (Section 4.2). They are grouped rather than random measurement splits: - DTI laboratories are defined by canonical compounds; - proteomic laboratories by compound groups; - cell laboratories by disjoint training interventions.

No compound or intervention is shared between laboratories. Grouping does not create real batch effects or protocol differences, however, and the revision says so.

To test the case you describe, the source-defined experiments assign a different released study to each laboratory (Section 4.5, Appendix B.2). The results are mixed, and we report all of them: - Proteomics: one study per laboratory (PTPC, Lin, Ruprecht and decryptE, with shared encoders and source-specific heads). For the revision we extended this study from one to eight seeds, with the analysis registered before the new fits. Target AUROC rises from 72.06 to 74.13 (paired +2.07, seed-block 95% CI [0.65, 3.38]; higher in seven of eight seeds). AP rises from 34.60 to 36.82 in mean (+2.22, CI [-1.19, 5.59]), but only four of eight seeds improve, so the AP gain is seed-dependent. Most of the benefit comes from decryptE; Lin and Ruprecht alone leave the target near its own level. - DTI (seed 61): adding Davis or Human raises target AUROC from 85.47 to 87.86 and 87.95, but BioSNAP lowers it by 3.03 and all three sources together by 1.18. - Cells (seeds 61-63): Jiang raises VCC Top-1 from 29.49 to 33.07, while Replogle, Nadig and all three sources lower it by 2.37, 2.55 and 2.15 points.

So the benefit of collaboration across real studies depends on their compatibility. With source-defined laboratories, no search policy beats the fixed design (Appendix E.4). Execution is simulated on shared compute, and we claim no formal privacy guarantee.

3. The pipeline's own "direct" search performs worse than a fixed model

"Direct" is configuration search without feedback from earlier evaluations: several candidates are trained and one is kept by development score. "Fixed" is the single default configuration. Both return one predictor; Section 4.2 now defines them.

With one laboratory, direct search, and every other search including BioCoLoop, selects on laboratory 0's small development panel. Its local-development rankings do not transfer reliably to the held-out set: single-laboratory BioCoLoop reaches 20.13 proteomics AP against 25.05 for the fixed model, and 14.15 Norman Top-1 against 25.56 (Appendix D.1). More laboratories enlarge both training data and the development population. Ten-laboratory BioCoLoop exceeds the single-laboratory fixed model on DTI, proteomics, VCC and Tahoe. On Norman, the single-laboratory fixed model (25.56) remains above it (22.25).

4. The baselines use one model while BioCoLoop uses ten laboratories

The ten laboratories hold data; they are not an ensemble. BioCoLoop, like every baseline, returns one predictor evaluated on the same test set. Table 1 compares developing that predictor with collaborators against developing it alone, the usual federated comparison. Its caption now says that rows at different access levels compare scenarios, not search methods.

Two controls hold access fixed: - A ten-laboratory fixed model (Appendix Table D.1). Shared data explain most of the gain on DTI, proteomics and VCC. BioCoLoop adds 1.96 AUROC on DTI and 6.77 Top-1 on Norman, nearly matches on proteomics, ties on VCC and is 3.48 lower on Tahoe. - A seed-42 pilot (Appendix C.4). Given the same ten laboratories, AI-Scientist-v2 matches BioCoLoop on four endpoints and exceeds it on DTI; repaired AI-Researcher matches on the three endpoints where it was scored.

5. "Asking an LLM to turn 4 knobs"

We do not claim a new optimizer over four hyperparameters, and the revision distinguishes BioCoLoop from federated hyperparameter search (including FedEx and evaluations using FedHPO-Bench) on this point. The contribution is a protocol and interface through which laboratories that cannot pool data jointly decide which predictor design to keep, what to try next and how to allocate further training, using only model updates and bounded per-laboratory development evidence. The same interface serves three task families and several research agents. The four-factor library is deliberately small so that the participation, proposal and allocation comparisons stay interpretable. Appendix E.9 widens it to 36 recipes that also vary the head, the loss and the schedule.

The clearest gain from collaborative decisions is allocation. At a fixed 800 aggregation rounds and 160 development evaluations, screening candidates on early collaborative evidence and training the promising ones longer raises Norman Top-1 from 11.48 to 22.32 (+10.84, 95% CI [4.54, 17.73]). Across 12 task-seed pairs it never loses to uniform allocation. Random promotion after the same screening performs comparably (6 wins, 6 losses), so the gain comes mainly from the screening-then-extend schedule (Appendix E.5).

Thank you again; these points led to the controls added in this revision.